

# Sparse Poisson GLM with Off-Diagonal Memory Kernel: Model, Inference, and Training

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## Abstract

Two-block inference scheme for a Poisson GLM with a lagged off-diagonal memory kernel on a non-stationary neuronal population. The model uses a diagonal lag-1 drive plus a shared history kernel on off-diagonal inputs. Alternating optimization yields tractable convex subproblems for the kernel ( $\lambda_2$  smoothness) and network parameters ( $\lambda_3$  off-diagonal sparsity).

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# 1 Generative Model

Population of  $N$  neurons,  $T$  time bins of width  $\Delta t$ . Spike counts  $Y_t \in \mathbb{N}_0^N$ ,  $t = 0, \dots, T-1$ .

**Parameters.** Diagonal:  $A^{\text{diag}} \in \mathbb{R}^N$  with  $(A^{\text{diag}})_i = (A_{\text{true}})_{ii}$ . Off-diagonal:  $A^{\text{off}} \in \mathbb{R}^{N \times N}$ , zeros on diagonal. Kernel:  $\kappa \in \mathbb{R}^M$ , 0-based indexing, anchor  $\kappa_0 = 1$ , learnable entries  $\kappa_1, \dots, \kappa_{M-1}$ . Bias:  $B \in \mathbb{R}^N$ . Clipping threshold:  $\eta_{\text{clip}} = 5$ .

**Off-diagonal history.**

$$S_t = \sum_{\ell=0}^{M-1} \kappa_\ell Y_{t-1-\ell} \in \mathbb{R}^N, \quad t-1-\ell \geq 0.$$

**State equation.**

$$\begin{aligned} \eta_{t,i} &= (A^{\text{diag}})_i Y_{t-1,i} + (A^{\text{off}} S_t)_i + B_i, \\ \mu_{t,i} &= \exp(\tilde{\eta}_{t,i}) \Delta t, \quad \tilde{\eta}_{t,i} = \min(\eta_{t,i}, \eta_{\text{clip}}). \end{aligned}$$

**Observation model.**

$$Y_{t,i} \sim \text{Poisson}(\mu_{t,i}), \quad i = 1, \dots, N, \quad t = 0, \dots, T-1.$$

## 2 Objective and Identifiability

**Log-likelihood.** Dropping the constant  $\log(Y_{t,i}!)$  terms:

$$\mathcal{L} = \sum_{i=1}^N \sum_{t=0}^{T-1} [Y_{t,i} \tilde{\eta}_{t,i} - \exp(\tilde{\eta}_{t,i}) \Delta t].$$

**Objective function.**

$$\mathcal{J} = \underbrace{\sum_{t=2}^T \sum_{i=1}^N [\mu_{t,i} - Y_{t,i} (\tilde{\eta}_{t,i} + \log \Delta t)]}_{\text{Poisson NLL}} + \underbrace{\lambda_2 \sum_{\ell=2}^{M-1} (\kappa_\ell - \kappa_{\ell-1})^2}_{\text{kernel smoothness (Block 1)}} + \underbrace{\lambda_3 \overline{A_{\text{off}}}}_{\text{off-diagonal sparsity (Block 2)}}, \quad (1)$$

where

$$\overline{A_{\text{off}}} = \frac{1}{N(N-1)} \sum_{\substack{i,j=1 \\ i \neq j}}^N |A_{ij}^{\text{off}}|. \quad (2)$$

**Identifiability.** The product  $A_{ij}^{\text{off}} \kappa_\ell$  has a scaling degeneracy, removed by the anchor  $\kappa_0 = 1$ . If clipping activates frequently, a censored Poisson likelihood should replace the standard one.

## 3 Inference: Two-Block Alternating Scheme

The bilinear structure  $A_{ij}^{\text{off}} \kappa_\ell$  precludes joint convexity. We use block coordinate descent: each block is convex given the other fixed, guaranteeing a non-increasing objective sequence converging to a stationary point.

### 3.1 Initialization

Initialize  $A^{\text{diag}}$  and  $B$  from mean firing rates and lag-1 covariances. Initialize  $\kappa$  with a damped-oscillator profile or simply  $\kappa_0 = 1$  with small subsequent entries.

### 3.2 Block 1 (kernel update)

Fix  $(A^{\text{diag}}, A^{\text{off}}, B)$ . Update  $\kappa_1, \dots, \kappa_{M-1}$  by minimizing  $\mathcal{J}$  with the  $\lambda_2$  smoothness penalty. The problem is convex in  $\kappa$  since  $S_t$  is linear in  $\kappa$ .

### 3.3 Block 2 (network update)

Fix  $\kappa$ . Update  $(A^{\text{diag}}, A^{\text{off}}, B)$  by minimizing the Poisson NLL plus the  $\lambda_3$  L1 penalty (2). Dale’s law sign constraints on columns of  $A^{\text{off}}$  can be added as linear inequalities.

## 4 Training Samples

Input: spike counts  $Y_t \in \mathbb{N}_0^N$ ,  $t = 0, \dots, T-1$ . Let  $M_{\text{cut}} \geq 2$  be the history cutoff.

**Sample construction.** Each sample at output time  $t_{\text{out}} \in [M_{\text{cut}}, T-1]$ :

$$X^{(t_{\text{out}})} = \{Y_{t_{\text{out}}-1}, Y_{t_{\text{out}}-2}, \dots, Y_{t_{\text{out}}-M_{\text{cut}}}\} \in \mathbb{N}_0^{N \times M_{\text{cut}}}, \quad y^{(t_{\text{out}})} = Y_{t_{\text{out}}} \in \mathbb{N}_0^N.$$

**Sampling strategy.** Draw  $t_{\text{out}}$  densely from  $[M_{\text{cut}}, T-1]$  with stride  $s$ ; optionally enforce a minimum separation to reduce autocorrelation. Total samples:

$$N_{\text{samples}} \approx \left\lfloor \frac{T - M_{\text{cut}}}{s} \right\rfloor.$$

For validation, sample windows within random non-overlapping blocks of length  $L \gg M_{\text{cut}}$  to obtain a temporally separated hold-out set.

## 5 Kernel Normalization

We use 0-based lag indexing  $\ell \in \{0, \dots, M-1\}$  with anchor  $\kappa_0 = 1$ . An alternative  $\|\kappa\|_1 = 1$  constraint was tested but significantly reduced non-stationarity in simulations, so the ground truth  $\kappa_{\text{true}}$  does not obey it.

The  $\lambda_2$  smoothness penalty imposes no parametric form on  $\kappa$ . If prior knowledge constrains the kernel shape, one can reparametrize  $\kappa$  with fewer parameters while keeping  $\kappa_0 = 1$ .

## 6 Implementation Notes

**Sparsity and sign constraints.** The  $\lambda_3$  penalty (2) promotes sparsity; normalization by  $N(N-1)$  makes the penalty strength network-size invariant. Column sign constraints (Dale’s law) can be imposed if neuron types are known.

**Regularization tuning.** Start  $\lambda_2$  small; increase if the kernel estimate oscillates. Tune  $\lambda_3$  to match expected connectivity density (e.g., 5–20%).